

Algebra 1

Unit 6

Lesson 5 Homework

Name Answer Key

Date _____

Period _____

The linear inequality $3x - 4y < 12$ is written in standard form.

1. Complete the statement.

The boundary line of this inequality is

o dotted

0 solid

because the points on the

boundary line are

o included in

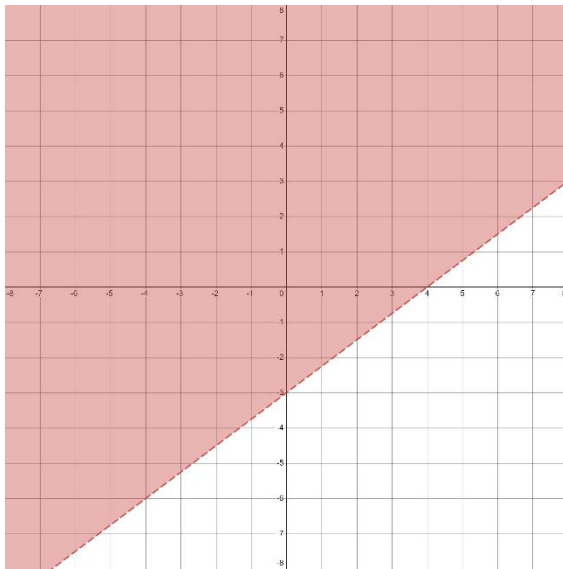
o excluded in

the set.

2. Write the equation of the boundary line of this inequality in standard form.

$$3x - 4y = 12$$

3. Graph the boundary line on the coordinate grid.



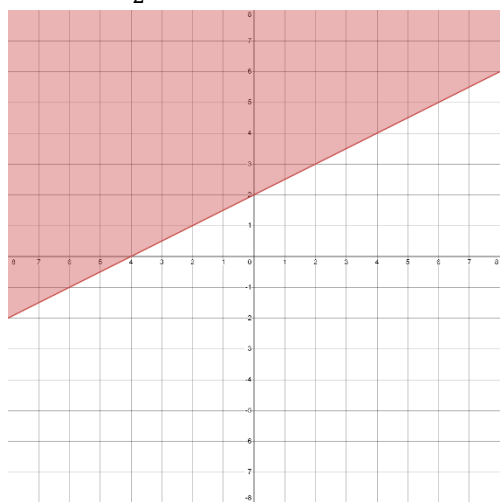
4. Explain how to determine which side of the boundary line to shade.

Choose a point, plug it in. If the inequality is true, shade in the same side as the point. If it's not true, shade in the opposite side.

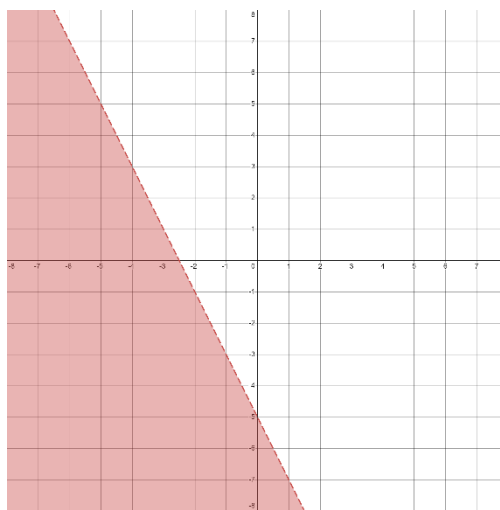
5. Shade the region on the graph that represents the solution set.

For each problem below, graph the given inequality.

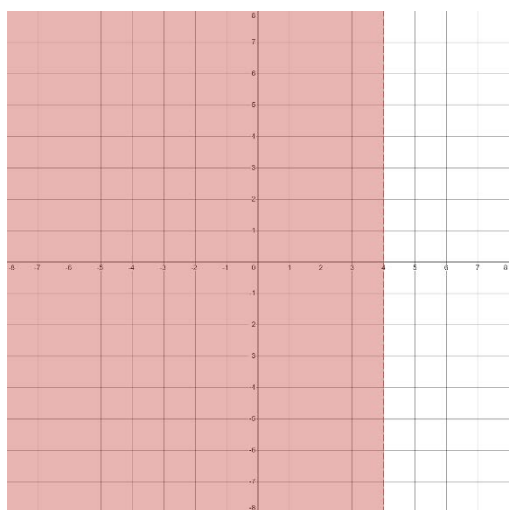
6. $y \geq \frac{1}{2}x + 2$



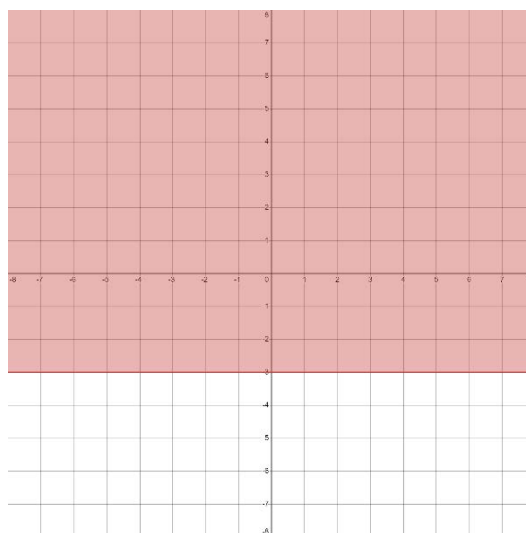
7. $y < -2x - 5$



8. $x < 4$



9. $y \geq -3$



10. $y - 1 \geq -3(x + 2) \rightarrow y \geq -3x - 5$

